

• Project / Task-3

Matrix Visualization complete code

MATRICES / NUMPY :

- Matrix is the tabular representation of the data
- Lot of datas are stored in table format,that is why Matrices is very very important topic in python
- As we working on dataframe so matrices are played a major rule
- List is one dimension & matrix is multidimension
- We will see tht & we gonna analyze the NBA players
- Hear i have taken top 10 highest paid player in 2015-2016 season
- We will analyze how 10 players have been playing over the past 10 years & we had the data for past 10yrs yrs
- Our main goal is to find trends,patterns & their performance for the past 10 yrs
- Ultimately they haven't always been top 10 player & lets see how they improving, what actually secreates or patterns
- Dont worry guys if you dont know anything about basket ball NBA
- I will explain indepth of everything
- lets analyze the statistics of the basket ball played
- gp - total games played,mpg - minutes per game,field goal(accuracy), ppg (points per game) -- this is no of point player has scores in that season
- Guys slowly i am bringing you into data analytics, jump into datavisualization using python
- I will give you the this code can everybody copy and paste your jupyter notebook
- Now i will explain with matrices

```
In [1]: #Import numpy
import numpy as np

#Seasons
Seasons = ["2010", "2011", "2012", "2013", "2014", "2015", "2016", "2017", "2018", "2019"]
Sdict = {"2010":0, "2011":1, "2012":2, "2013":3, "2014":4, "2015":5, "2016":6, "2017":7

#Players
Players = ["Sachin", "Rahul", "Smith", "Sami", "Pollard", "Morris", "Samson", "Dhoni", "
Pdict = {"Sachin":0, "Rahul":1, "Smith":2, "Sami":3, "Pollard":4, "Morris":5, "Samson"

#Salaries
Sachin_Salary = [15946875, 17718750, 19490625, 21262500, 23034375, 24806250, 25244493,
```

```

Rahul_Salary = [12000000,12744189,13488377,14232567,14976754,16324500,18038573,1
Smith_Salary = [4621800,5828090,13041250,14410581,15779912,14500000,16022500,175
Sami_Salary = [3713640,4694041,13041250,14410581,15779912,17149243,18518574,1945
Pollard_Salary = [4493160,4806720,6061274,13758000,15202590,16647180,18091770,19
Morris_Salary = [3348000,4235220,12455000,14410581,15779912,14500000,16022500,17
Samson_Salary = [3144240,3380160,3615960,4574189,13520500,14940153,16359805,1777
Dhoni_Salary = [0,0,4171200,4484040,4796880,6053663,15506632,16669630,17832627,1
Kohli_Salary = [0,0,0,4822800,5184480,5546160,6993708,16402500,17632688,18862875
Sky_Salary = [3031920,3841443,13041250,14410581,15779912,14200000,15691000,17182
#Matrix
Salary = np.array([Sachin_Salary, Rahul_Salary, Smith_Salary, Sami_Salary, Polla

#Games
Sachin_G = [80,77,82,82,73,82,58,78,6,35]
Rahul_G = [82,57,82,79,76,72,60,72,79,80]
Smith_G = [79,78,75,81,76,79,62,76,77,69]
Sami_G = [80,65,77,66,69,77,55,67,77,40]
Pollard_G = [82,82,82,79,82,78,54,76,71,41]
Morris_G = [70,69,67,77,70,77,57,74,79,44]
Samson_G = [78,64,80,78,45,80,60,70,62,82]
Dhoni_G = [35,35,80,74,82,78,66,81,81,27]
Kohli_G = [40,40,40,81,78,81,39,0,10,51]
Sky_G = [75,51,51,79,77,76,49,69,54,62]
#Matrix
Games = np.array([Sachin_G, Rahul_G, Smith_G, Sami_G, Pollard_G, Morris_G, Samso

#Points
Sachin_PTS = [2832,2430,2323,2201,1970,2078,1616,2133,83,782]
Rahul_PTS = [1653,1426,1779,1688,1619,1312,1129,1170,1245,1154]
Smith_PTS = [2478,2132,2250,2304,2258,2111,1683,2036,2089,1743]
Sami_PTS = [2122,1881,1978,1504,1943,1970,1245,1920,2112,966]
Pollard_PTS = [1292,1443,1695,1624,1503,1784,1113,1296,1297,646]
Morris_PTS = [1572,1561,1496,1746,1678,1438,1025,1232,1281,928]
Samson_PTS = [1258,1104,1684,1781,841,1268,1189,1186,1185,1564]
Dhoni_PTS = [903,903,1624,1871,2472,2161,1850,2280,2593,686]
Kohli_PTS = [597,597,597,1361,1619,2026,852,0,159,904]
Sky_PTS = [2040,1397,1254,2386,2045,1941,1082,1463,1028,1331]
#Matrix
Points = np.array([Sachin_PTS, Rahul_PTS, Smith_PTS, Sami_PTS, Pollard_PTS, Morr

```

In [142...

Salary # martrix format

```
Out[142... array([[15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
        25244493, 27849149, 30453805, 23500000],
       [12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
        18038573, 19752645, 21466718, 23180790],
       [ 4621800,  5828090, 13041250, 14410581, 15779912, 14500000,
        16022500, 17545000, 19067500, 20644400],
       [ 3713640,  4694041, 13041250, 14410581, 15779912, 17149243,
        18518574, 19450000, 22407474, 22458000],
       [ 4493160,  4806720,  6061274, 13758000, 15202590, 16647180,
        18091770, 19536360, 20513178, 21436271],
       [ 3348000,  4235220, 12455000, 14410581, 15779912, 14500000,
        16022500, 17545000, 19067500, 20644400],
       [ 3144240,  3380160,  3615960,  4574189, 13520500, 14940153,
        16359805, 17779458, 18668431, 20068563],
       [      0,      0,  4171200,  4484040,  4796880,  6053663,
        15506632, 16669630, 17832627, 18995624],
       [      0,      0,      0,  4822800,  5184480,  5546160,
        6993708, 16402500, 17632688, 18862875],
       [ 3031920,  3841443, 13041250, 14410581, 15779912, 14200000,
        15691000, 17182000, 18673000, 15000000]])
```

```
In [3]: # Building your first matrix -
        Games
```

```
Out[3]: array([[80, 77, 82, 82, 73, 82, 58, 78,  6, 35],
       [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
       [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
       [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
       [82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
       [70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
       [78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
       [35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
       [40, 40, 40, 81, 78, 81, 39,  0, 10, 51],
       [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]])
```

```
In [5]: Points
```

```
Out[5]: array([[2832, 2430, 2323, 2201, 1970, 2078, 1616, 2133,  83, 782],
       [1653, 1426, 1779, 1688, 1619, 1312, 1129, 1170, 1245, 1154],
       [2478, 2132, 2250, 2304, 2258, 2111, 1683, 2036, 2089, 1743],
       [2122, 1881, 1978, 1504, 1943, 1970, 1245, 1920, 2112,  966],
       [1292, 1443, 1695, 1624, 1503, 1784, 1113, 1296, 1297,  646],
       [1572, 1561, 1496, 1746, 1678, 1438, 1025, 1232, 1281,  928],
       [1258, 1104, 1684, 1781,  841, 1268, 1189, 1186, 1185, 1564],
       [ 903,  903, 1624, 1871, 2472, 2161, 1850, 2280, 2593,  686],
       [ 597,  597,  597, 1361, 1619, 2026,  852,  0, 159,  904],
       [2040, 1397, 1254, 2386, 2045, 1941, 1082, 1463, 1028, 1331]])
```

```
In [7]: mydata = np.arange(0,20)
        print(mydata)
```

```
[ 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18 19]
```

```
In [8]: np.reshape(mydata,(4,5)) # 5 rows & 4 columns
```

```
Out[8]: array([[ 0,  1,  2,  3,  4],
       [ 5,  6,  7,  8,  9],
       [10, 11, 12, 13, 14],
       [15, 16, 17, 18, 19]])
```

```
In [9]: mydata
```

```
Out[9]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
               17, 18, 19])
```

```
In [10]: #np.reshape(mydata,(5,4), order = 'c') #'C' means to read / write the elements u
MATR1 = np.reshape(mydata, (5,4), order = 'c')
MATR1      # Here C= Row wise ,F = column wise , A = Randomly wise .
```

```
Out[10]: array([[ 0,  1,  2,  3],
                [ 4,  5,  6,  7],
                [ 8,  9, 10, 11],
                [12, 13, 14, 15],
                [16, 17, 18, 19]])
```

```
In [11]: MATR1
```

```
Out[11]: array([[ 0,  1,  2,  3],
                [ 4,  5,  6,  7],
                [ 8,  9, 10, 11],
                [12, 13, 14, 15],
                [16, 17, 18, 19]])
```

```
In [12]: MATR1[4,3]
```

```
Out[12]: np.int64(19)
```

```
In [13]: MATR1[3,3]
```

```
Out[13]: np.int64(15)
```

```
In [14]: MATR1
```

```
Out[14]: array([[ 0,  1,  2,  3],
                [ 4,  5,  6,  7],
                [ 8,  9, 10, 11],
                [12, 13, 14, 15],
                [16, 17, 18, 19]])
```

```
In [16]: MATR1[-3,-2]
```

```
Out[16]: np.int64(10)
```

```
In [17]: MATR1
```

```
Out[17]: array([[ 0,  1,  2,  3],
                [ 4,  5,  6,  7],
                [ 8,  9, 10, 11],
                [12, 13, 14, 15],
                [16, 17, 18, 19]])
```

```
In [18]: mydata
```

```
Out[18]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
               17, 18, 19])
```

```
In [19]: MATR2 = np.reshape(mydata, (5,4), order = 'F') # reshape behaviour are - 'C', 'F'
MATR2
```

```
Out[19]: array([[ 0,  5, 10, 15],
                [ 1,  6, 11, 16],
                [ 2,  7, 12, 17],
                [ 3,  8, 13, 18],
                [ 4,  9, 14, 19]])
```

```
In [20]: MATR2[4,3]
```

```
Out[20]: np.int64(19)
```

```
In [22]: MATR2[0:2]
```

```
Out[22]: array([[ 0,  5, 10, 15],
                [ 1,  6, 11, 16]])
```

```
In [23]: MATR2[-1,-2]
```

```
Out[23]: np.int64(14)
```

```
In [24]: mydata
```

```
Out[24]: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
                17, 18, 19])
```

```
In [101... MATR3 = np.reshape(mydata, (5,4), order = 'A')
MATR3
```

```
Out[101... array([[ 0,  1,  2,  3],
                [ 4,  5,  6,  7],
                [ 8,  9, 10, 11],
                [12, 13, 14, 15],
                [16, 17, 18, 19]])
```

```
In [26]: a1 = ['welcome', 'to', 'datascience']
a2 = ['required', 'hard', 'work' ]
a3 = [1,2,3]
```

```
In [28]: [a1, a2, a3]
```

```
Out[28]: [['welcome', 'to', 'datascience'], ['required', 'hard', 'work'], [1, 2, 3]]
```

```
In [29]: np.array([a1,a2,a3])
```

```
Out[29]: array(['welcome', 'to', 'datascience'],
                ['required', 'hard', 'work'],
                ['1', '2', '3']], dtype='<U21')
```

```
In [31]: Games
```

```
Out[31]: array([[80, 77, 82, 82, 73, 82, 58, 78, 6, 35],
               [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
               [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
               [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
               [82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
               [70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
               [78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
               [35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
               [40, 40, 40, 81, 78, 81, 39, 0, 10, 51],
               [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]])
```

```
In [32]: Games[5]
```

```
Out[32]: array([70, 69, 67, 77, 70, 77, 57, 74, 79, 44])
```

```
In [33]: Games[0:4]
```

```
Out[33]: array([[80, 77, 82, 82, 73, 82, 58, 78, 6, 35],
               [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
               [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
               [80, 65, 77, 66, 69, 77, 55, 67, 77, 40]])
```

```
In [35]: Games[-3, -2]
```

```
Out[35]: np.int64(81)
```

```
In [36]: Points
```

```
Out[36]: array([[2832, 2430, 2323, 2201, 1970, 2078, 1616, 2133, 83, 782],
               [1653, 1426, 1779, 1688, 1619, 1312, 1129, 1170, 1245, 1154],
               [2478, 2132, 2250, 2304, 2258, 2111, 1683, 2036, 2089, 1743],
               [2122, 1881, 1978, 1504, 1943, 1970, 1245, 1920, 2112, 966],
               [1292, 1443, 1695, 1624, 1503, 1784, 1113, 1296, 1297, 646],
               [1572, 1561, 1496, 1746, 1678, 1438, 1025, 1232, 1281, 928],
               [1258, 1104, 1684, 1781, 841, 1268, 1189, 1186, 1185, 1564],
               [903, 903, 1624, 1871, 2472, 2161, 1850, 2280, 2593, 686],
               [597, 597, 597, 1361, 1619, 2026, 852, 0, 159, 904],
               [2040, 1397, 1254, 2386, 2045, 1941, 1082, 1463, 1028, 1331]])
```

```
In [37]: Points[0]
```

```
Out[37]: array([2832, 2430, 2323, 2201, 1970, 2078, 1616, 2133, 83, 782])
```

```
In [38]: Points[7, 2]
```

```
Out[38]: np.int64(1624)
```

```
In [39]: Points[3:5]
```

```
Out[39]: array([[2122, 1881, 1978, 1504, 1943, 1970, 1245, 1920, 2112, 966],
               [1292, 1443, 1695, 1624, 1503, 1784, 1113, 1296, 1297, 646]])
```

```
In [40]: Points[-6, -2]
```

```
Out[40]: np.int64(1297)
```

===== DICTIONARY =====

dict does not maintain the order

```
In [41]: dict1 = {'key1':'val1', 'key2':'val2', 'key3':'val3'}
```

```
In [42]: dict1
```

```
Out[42]: {'key1': 'val1', 'key2': 'val2', 'key3': 'val3'}
```

```
In [43]: dict1['key2']
```

```
Out[43]: 'val2'
```

```
In [44]: dict2 = {'bang':2,'hyd':'we are hear', 'pune':True}
```

```
In [45]: dict2
```

```
Out[45]: {'bang': 2, 'hyd': 'we are hear', 'pune': True}
```

```
In [46]: dict3 = {'Germany':'I have been here', 'France':2, 'Spain': True}
```

```
In [47]: dict3
```

```
Out[47]: {'Germany': 'I have been here', 'France': 2, 'Spain': True}
```

```
In [48]: dict3['Germany']
```

```
Out[48]: 'I have been here'
```

```
In [49]: Games
```

```
Out[49]: array([[80, 77, 82, 82, 73, 82, 58, 78, 6, 35],
                [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
                [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
                [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
                [82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
                [70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
                [78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
                [35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
                [40, 40, 40, 81, 78, 81, 39, 0, 10, 51],
                [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]])
```

```
In [50]: Pdict
```

```
Out[50]: {'Sachin': 0,
          'Rahul': 1,
          'Smith': 2,
          'Sami': 3,
          'Pollard': 4,
          'Morris': 5,
          'Samson': 6,
          'Dhoni': 7,
          'Kohli': 8,
          'Sky': 9}
```

```
In [51]: # how do i know player kobe bryant is at
```

```
Pdict['Sachin']
```

```
Out[51]: 0
```

```
In [52]: Games[0]
```

```
Out[52]: array([80, 77, 82, 82, 73, 82, 58, 78,  6, 35])
```

```
In [53]: Games[1]
```

```
Out[53]: array([82, 57, 82, 79, 76, 72, 60, 72, 79, 80])
```

```
In [55]: Pdict['Rahul']
```

```
Out[55]: 1
```

Games

```
In [56]: Games[Pdict['Rahul']]
```

```
Out[56]: array([82, 57, 82, 79, 76, 72, 60, 72, 79, 80])
```

```
In [58]: Salary
```

```
Out[58]: array([[15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
                25244493, 27849149, 30453805, 23500000],
                [12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
                18038573, 19752645, 21466718, 23180790],
                [ 4621800,  5828090, 13041250, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
                [ 3713640,  4694041, 13041250, 14410581, 15779912, 17149243,
                18518574, 19450000, 22407474, 22458000],
                [ 4493160,  4806720,  6061274, 13758000, 15202590, 16647180,
                18091770, 19536360, 20513178, 21436271],
                [ 3348000,  4235220, 12455000, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
                [ 3144240,  3380160,  3615960,  4574189, 13520500, 14940153,
                16359805, 17779458, 18668431, 20068563],
                [      0,      0,  4171200,  4484040,  4796880,  6053663,
                15506632, 16669630, 17832627, 18995624],
                [      0,      0,      0,  4822800,  5184480,  5546160,
                6993708, 16402500, 17632688, 18862875],
                [ 3031920,  3841443, 13041250, 14410581, 15779912, 14200000,
                15691000, 17182000, 18673000, 15000000]])
```

```
In [59]: Salary[2,5]
```

```
Out[59]: np.int64(14500000)
```

```
In [60]: Salary[Pdict['Sky']][Sdict['2019']]
```

```
Out[60]: np.int64(15000000)
```


In [67]: `Salary/Games` # Here Warning show with the red sell below:

C:\Users\Lenovo\AppData\Local\Temp\ipykernel_6340\3709746658.py:1: RuntimeWarning: divide by zero encountered in divide
Salary/Games

```
Out[67]: array([[ 199335.9375    ,  230113.63636364,  237690.54878049,
  259298.7804878 ,  315539.38356164,  302515.24390244,
  435249.87931034,  357040.37179487,  5075634.16666667,
  671428.57142857],
 [ 146341.46341463,  223582.26315789,  164492.40243902,
  180159.07594937,  197062.55263158,  226729.16666667,
  300642.88333333,  274342.29166667,  271730.60759494,
  289759.875     ],
 [  58503.79746835,   74719.1025641 ,  173883.33333333,
  177908.40740741,  207630.42105263,  183544.30379747,
  258427.41935484,  230855.26315789,  247629.87012987,
  299194.20289855],
 [  46420.5      ,   72216.01538462,  169366.88311688,
  218342.13636364,  228694.37681159,  222717.44155844,
  336701.34545455,  290298.50746269,  291006.15584416,
  561450.         ],
 [  54794.63414634,   58618.53658537,   73917.97560976,
  174151.89873418,  185397.43902439,  213425.38461538,
  335032.77777778,  257057.36842105,  288918.         ,
  522835.87804878],
 [  47828.57142857,   61380.         ,  185895.52238806,
  187150.4025974 ,  225427.31428571,  188311.68831169,
  281096.49122807,  237094.59459459,  241360.75949367,
  469190.90909091],
 [  40310.76923077,   52815.         ,   45199.5      ,
   58643.44871795,  300455.55555556,  186751.9125     ,
  272663.41666667,  253992.25714286,  301103.72580645,
  244738.57317073],
 [    0.         ,    0.         ,   52140.         ,
   60595.13513514,   58498.53658537,   77611.06410256,
  234948.96969697,  205797.90123457,  220155.88888889,
   703541.62962963],
 [    0.         ,    0.         ,    0.         ,
   59540.74074074,   66467.69230769,   68471.11111111,
  179325.84615385,                inf,  1763268.8      ,
   369860.29411765],
 [  40425.6      ,   75322.41176471,  255710.78431373,
  182412.41772152,  204933.92207792,  186842.10526316,
  320224.48979592,  249014.49275362,  345796.2962963 ,
  241935.48387097]])
```

In [68]: `np.round(Salary/Games)`

C:\Users\Lenovo\AppData\Local\Temp\ipykernel_6340\2909567671.py:1: RuntimeWarning: divide by zero encountered in divide
np.round(Salary/Games)

```
Out[68]: array([[ 199336.,  230114.,  237691.,  259299.,  315539.,  302515.,
    435250.,  357040.,  5075634.,  671429.],
 [ 146341.,  223582.,  164492.,  180159.,  197063.,  226729.,
    300643.,  274342.,  271731.,  289760.],
 [  58504.,   74719.,  173883.,  177908.,  207630.,  183544.,
    258427.,  230855.,  247630.,  299194.],
 [  46420.,   72216.,  169367.,  218342.,  228694.,  222717.,
    336701.,  290299.,  291006.,  561450.],
 [  54795.,   58619.,   73918.,  174152.,  185397.,  213425.,
    335033.,  257057.,  288918.,  522836.],
 [  47829.,   61380.,  185896.,  187150.,  225427.,  188312.,
    281096.,  237095.,  241361.,  469191.],
 [  40311.,   52815.,   45200.,   58643.,  300456.,  186752.,
    272663.,  253992.,  301104.,  244739.],
 [      0.,      0.,   52140.,   60595.,   58499.,   77611.,
    234949.,  205798.,  220156.,  703542.],
 [      0.,      0.,      0.,   59541.,   66468.,   68471.,
    179326.,    inf,  1763269.,  369860.],
 [  40426.,   75322.,  255711.,  182412.,  204934.,  186842.,
    320224.,  249014.,  345796.,  241935.]])
```

```
In [69]: import warnings
warnings.filterwarnings('ignore')
#np.round(FieldGoals/Games)
#FieldGoals/Games # this matrix is lot of decimal points yo can not round
#round()
```

--- First visualization ----

```
In [70]: import numpy as np
import matplotlib.pyplot as plt
```

```
In [71]: %matplotlib inline
```

```
In [72]: Salary
```

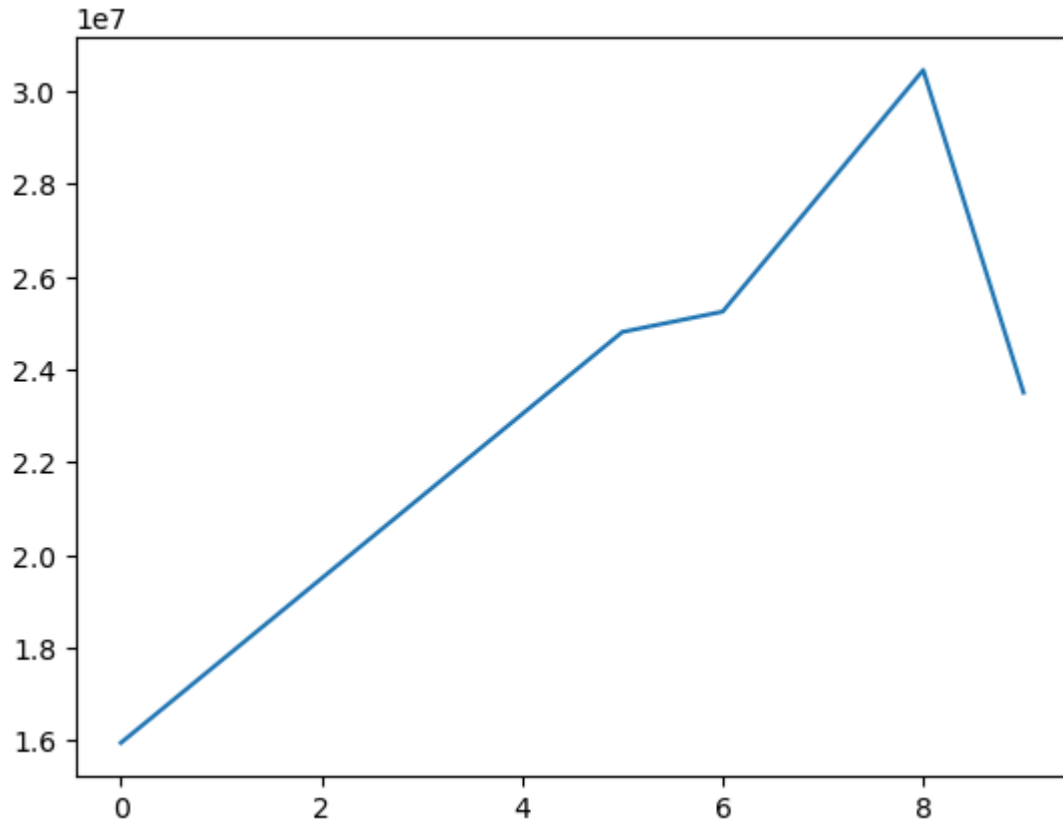
```
Out[72]: array([[15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
    25244493, 27849149, 30453805, 23500000],
 [12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
    18038573, 19752645, 21466718, 23180790],
 [ 4621800,  5828090, 13041250, 14410581, 15779912, 14500000,
    16022500, 17545000, 19067500, 20644400],
 [ 3713640,  4694041, 13041250, 14410581, 15779912, 17149243,
    18518574, 19450000, 22407474, 22458000],
 [ 4493160,  4806720,  6061274, 13758000, 15202590, 16647180,
    18091770, 19536360, 20513178, 21436271],
 [ 3348000,  4235220, 12455000, 14410581, 15779912, 14500000,
    16022500, 17545000, 19067500, 20644400],
 [ 3144240,  3380160,  3615960,  4574189, 13520500, 14940153,
    16359805, 17779458, 18668431, 20068563],
 [      0,      0,  4171200,  4484040,  4796880,  6053663,
    15506632, 16669630, 17832627, 18995624],
 [      0,      0,      0,  4822800,  5184480,  5546160,
    6993708, 16402500, 17632688, 18862875],
 [ 3031920,  3841443, 13041250, 14410581, 15779912, 14200000,
    15691000, 17182000, 18673000, 15000000]])
```

```
In [73]: Salary[0]
```

```
Out[73]: array([15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
                25244493, 27849149, 30453805, 23500000])
```

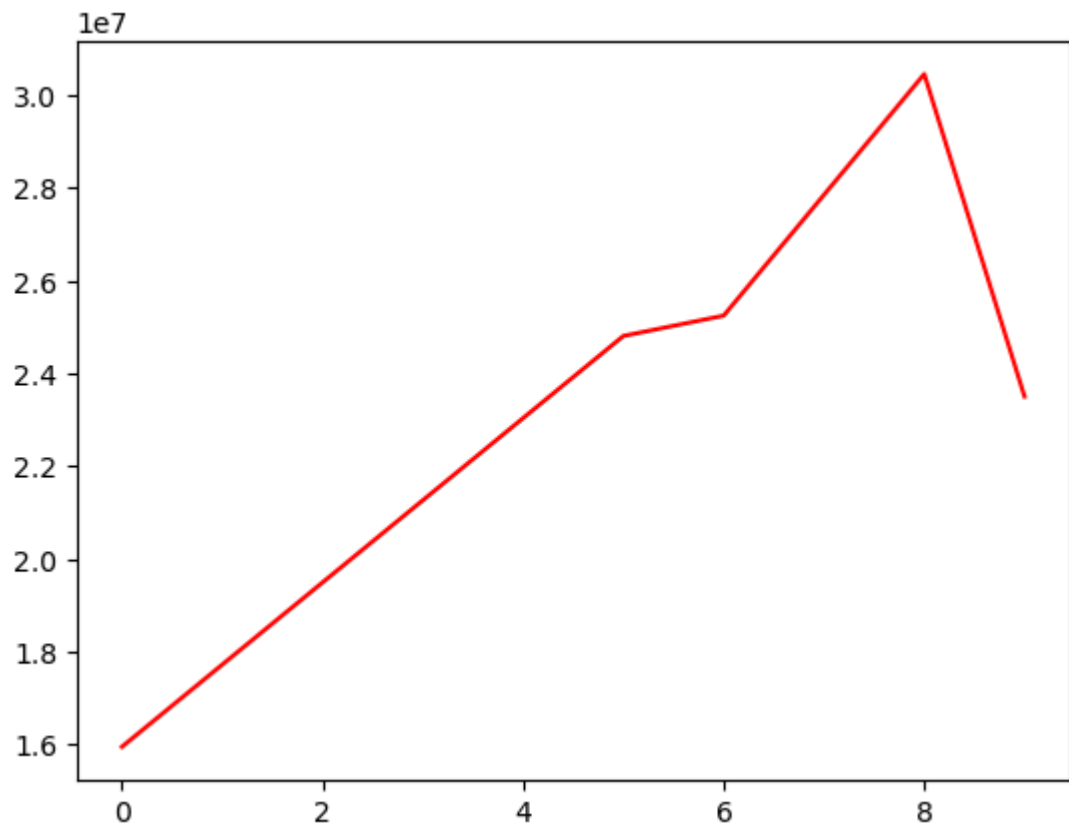
```
In [74]: plt.plot(Salary[0])
```

```
Out[74]: [<matplotlib.lines.Line2D at 0x1ef3bec0b90>]
```



```
In [75]: plt.plot(Salary[0], c='red') # Changing color of line graph by using color cod
```

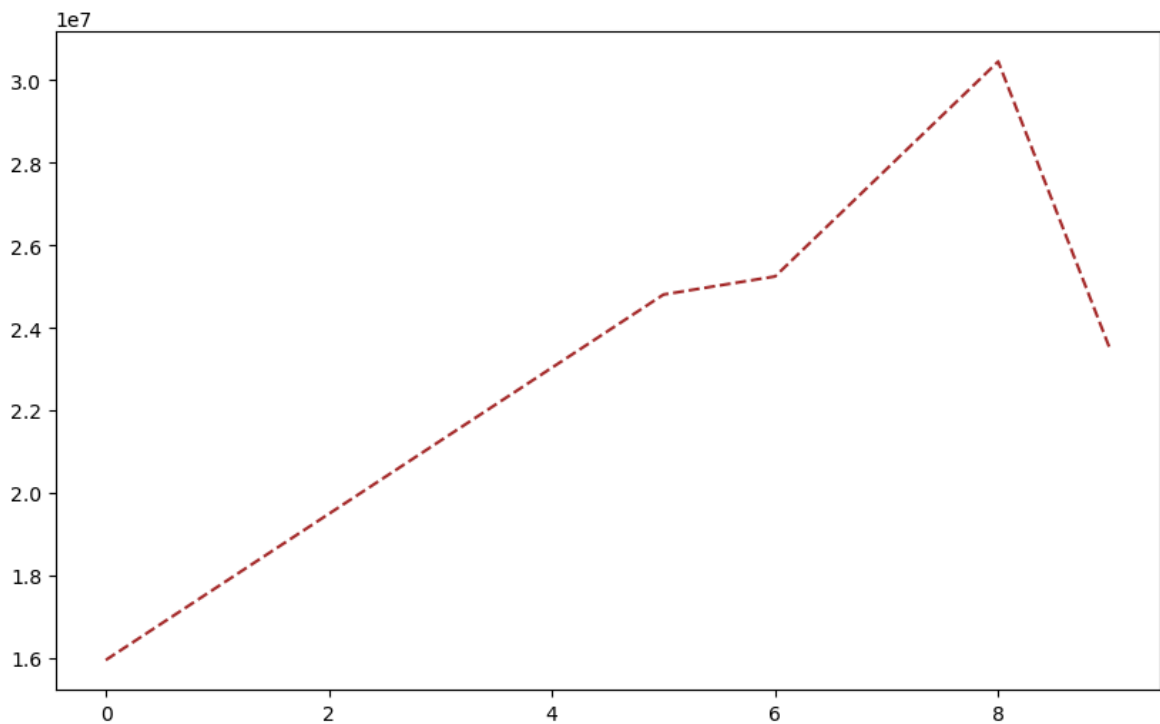
```
Out[75]: [<matplotlib.lines.Line2D at 0x1ef3bfa1310>]
```



```
In [77]: %matplotlib inline
plt.rcParams['figure.figsize'] = 10,6
```

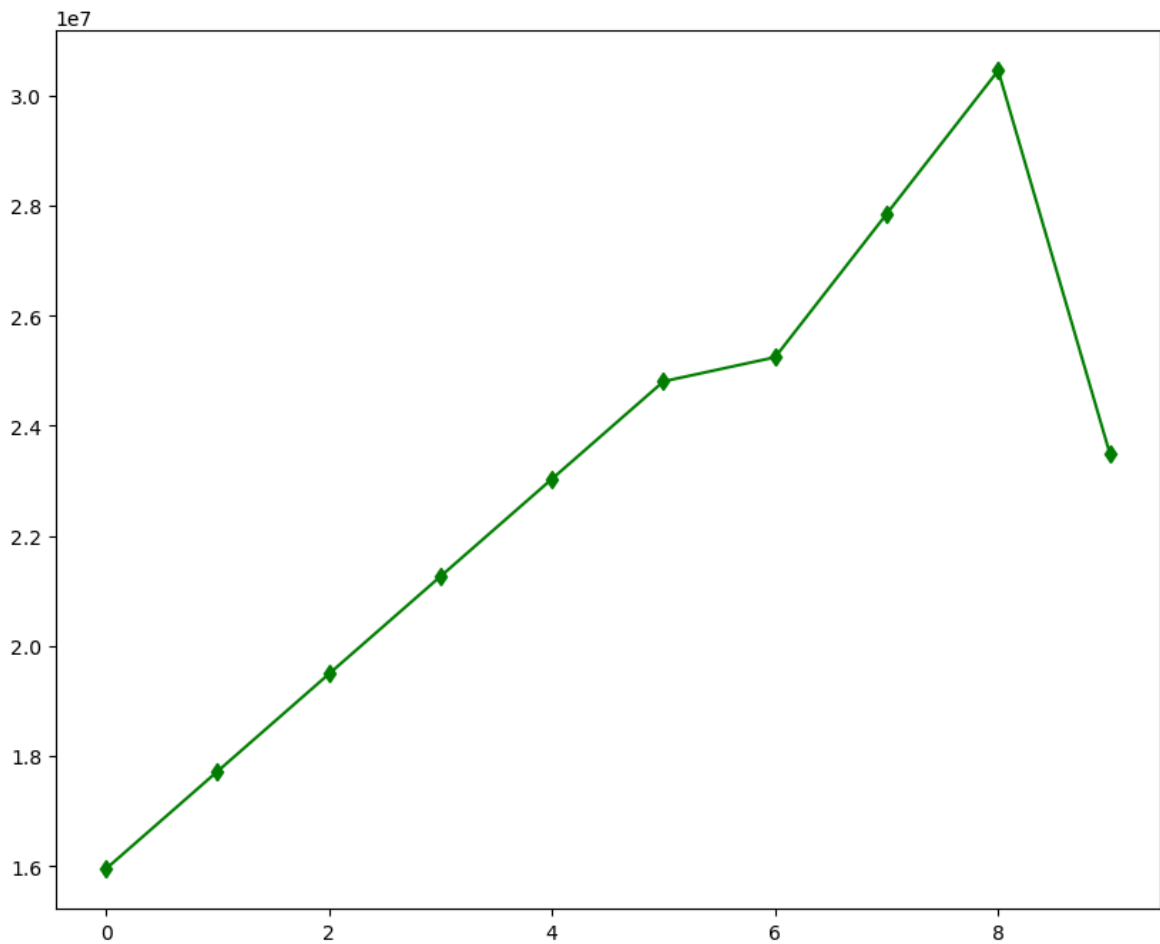
```
In [78]: plt.plot(Salary[0], c = 'Brown', ls = '--')
```

```
Out[78]: [<matplotlib.lines.Line2D at 0x1ef3c0302d0>]
```



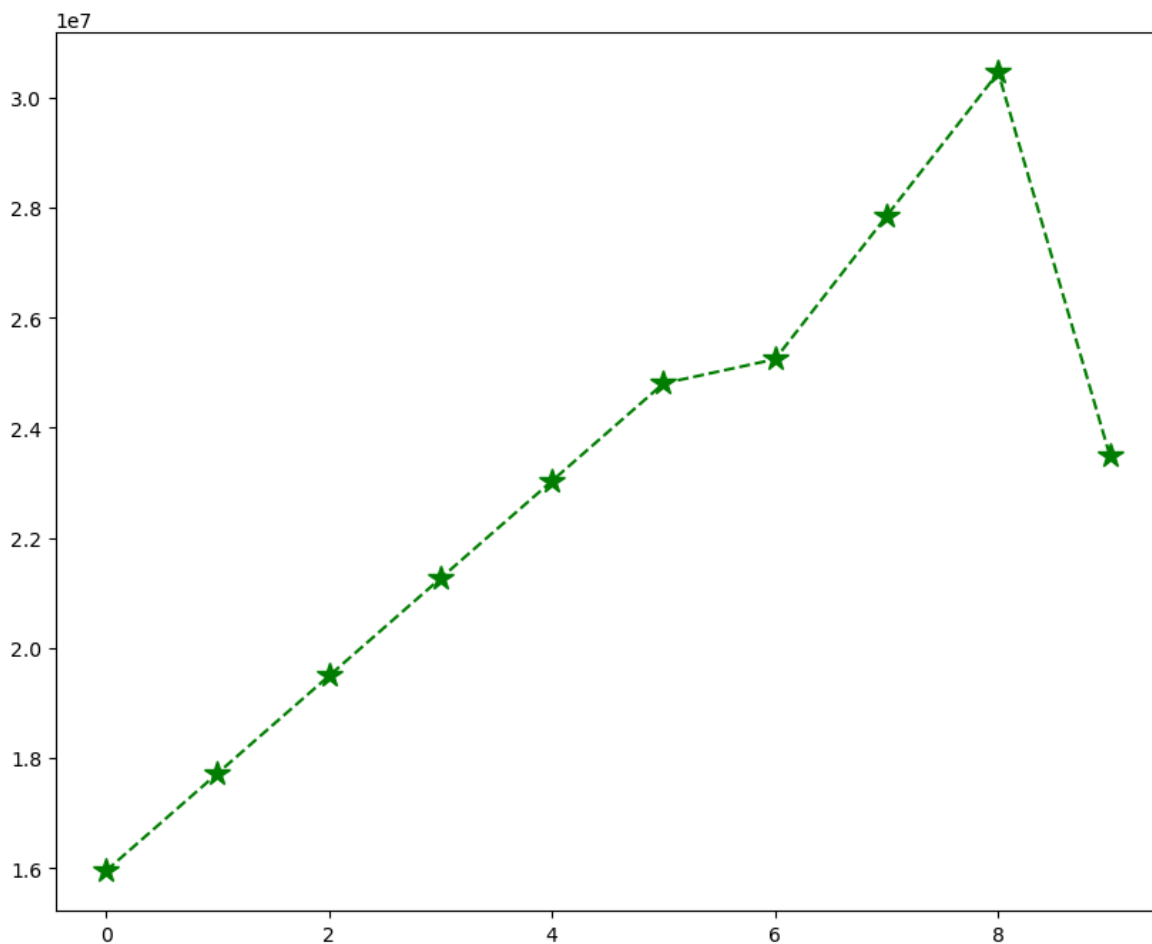
```
In [141... plt.plot(Salary[0], c='Green', ls = '-', marker = 'd') # s - squares
```

```
Out[141... [<matplotlib.lines.Line2D at 0x1ef423820d0>]
```



```
In [80]: %matplotlib inline
plt.rcParams['figure.figsize'] = 10,8 #runtime configuration parameter
```

```
In [139... plt.plot(Salary[0], c='Green', ls = '--', marker = '*', ms = 13)
plt.show()
```



```
In [82]: list(range(0,10))
```

```
Out[82]: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

```
In [83]: Sdict
```

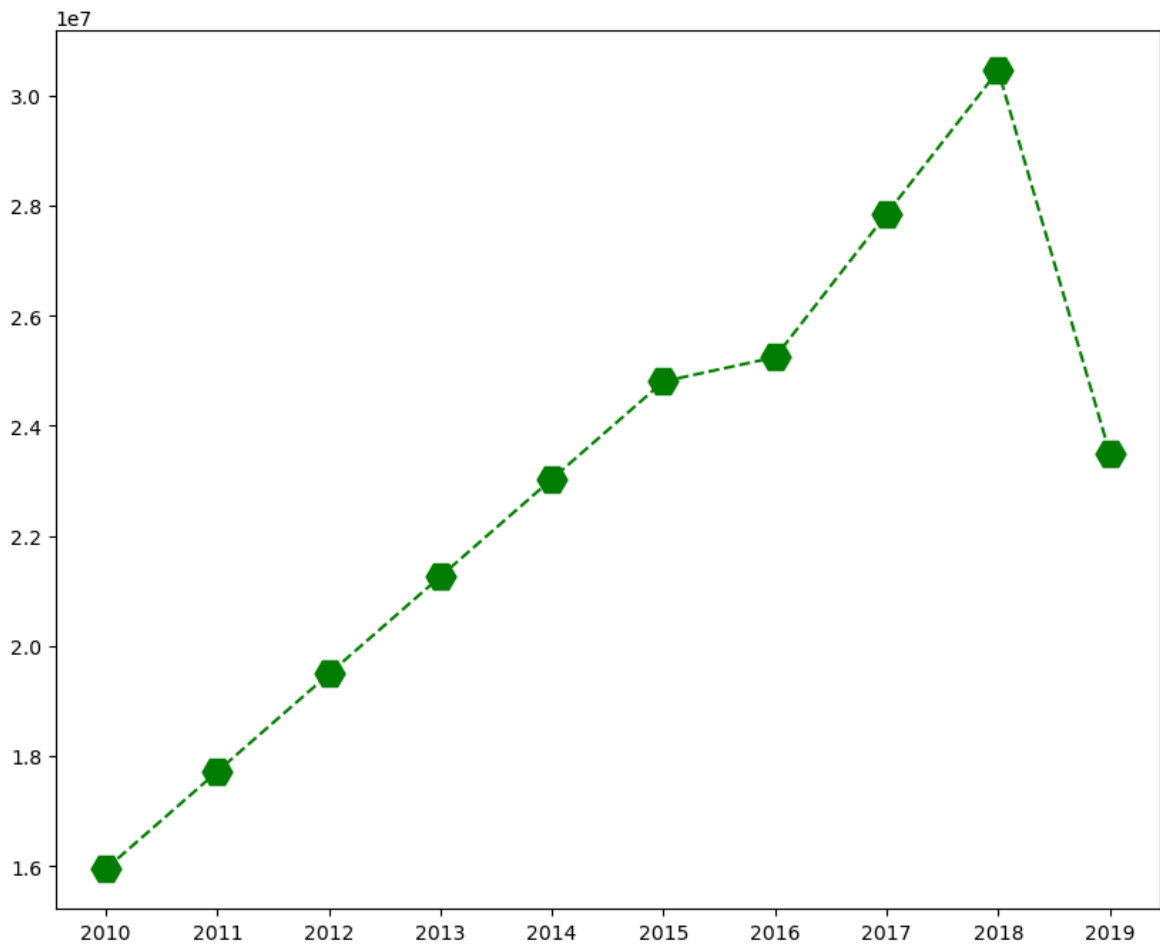
```
Out[83]: {'2010': 0,  
          '2011': 1,  
          '2012': 2,  
          '2013': 3,  
          '2014': 4,  
          '2015': 5,  
          '2016': 6,  
          '2017': 7,  
          '2018': 8,  
          '2019': 9}
```

```
In [84]: Pdict
```

```
Out[84]: {'Sachin': 0,  
          'Rahul': 1,  
          'Smith': 2,  
          'Sami': 3,  
          'Pollard': 4,  
          'Morris': 5,  
          'Samson': 6,  
          'Dhoni': 7,  
          'Kohli': 8,  
          'Sky': 9}
```

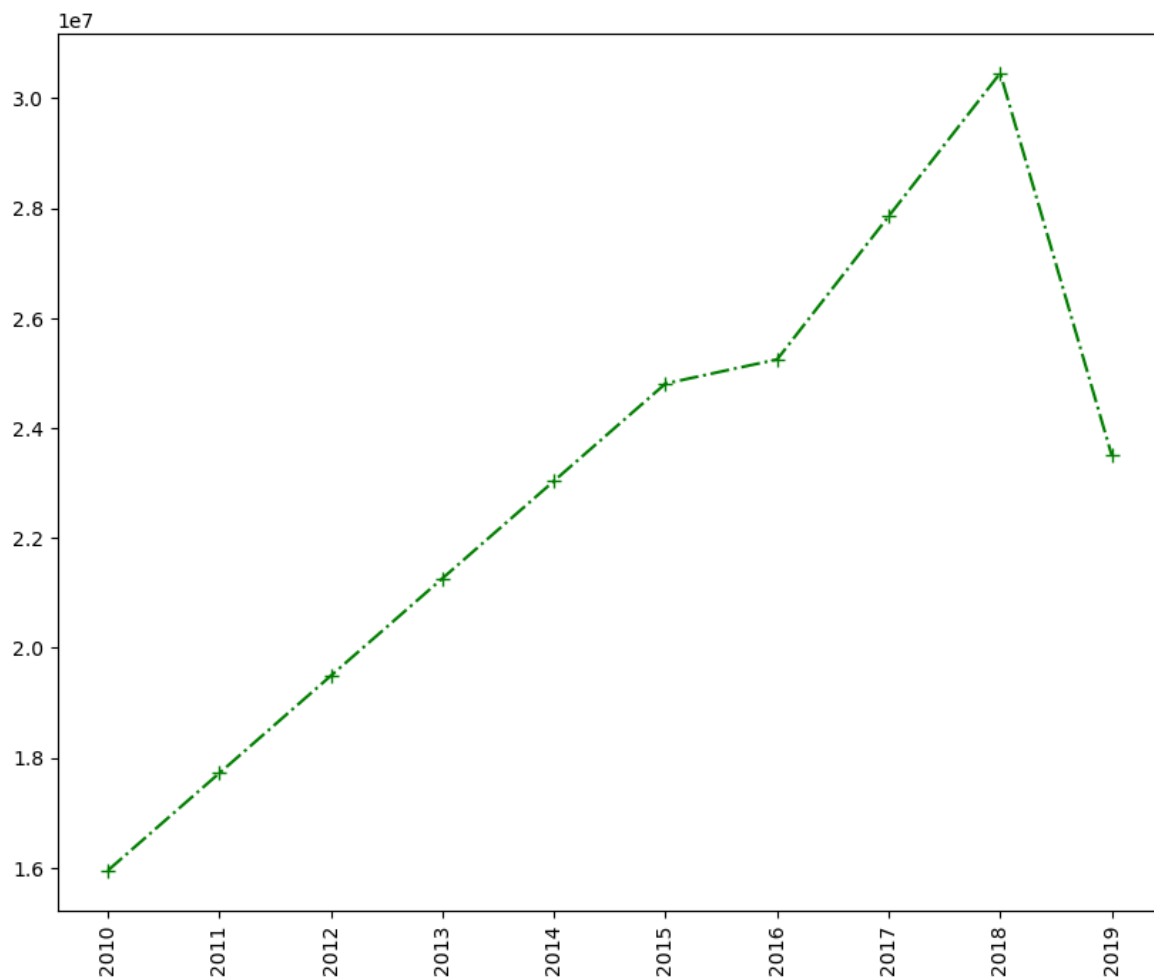
In [137...

```
plt.plot(Salary[0], c='Green', ls = '--', marker = 'H', ms = 15)  
plt.xticks(list(range(0,10)), Seasons)  
plt.show()
```



In [133...

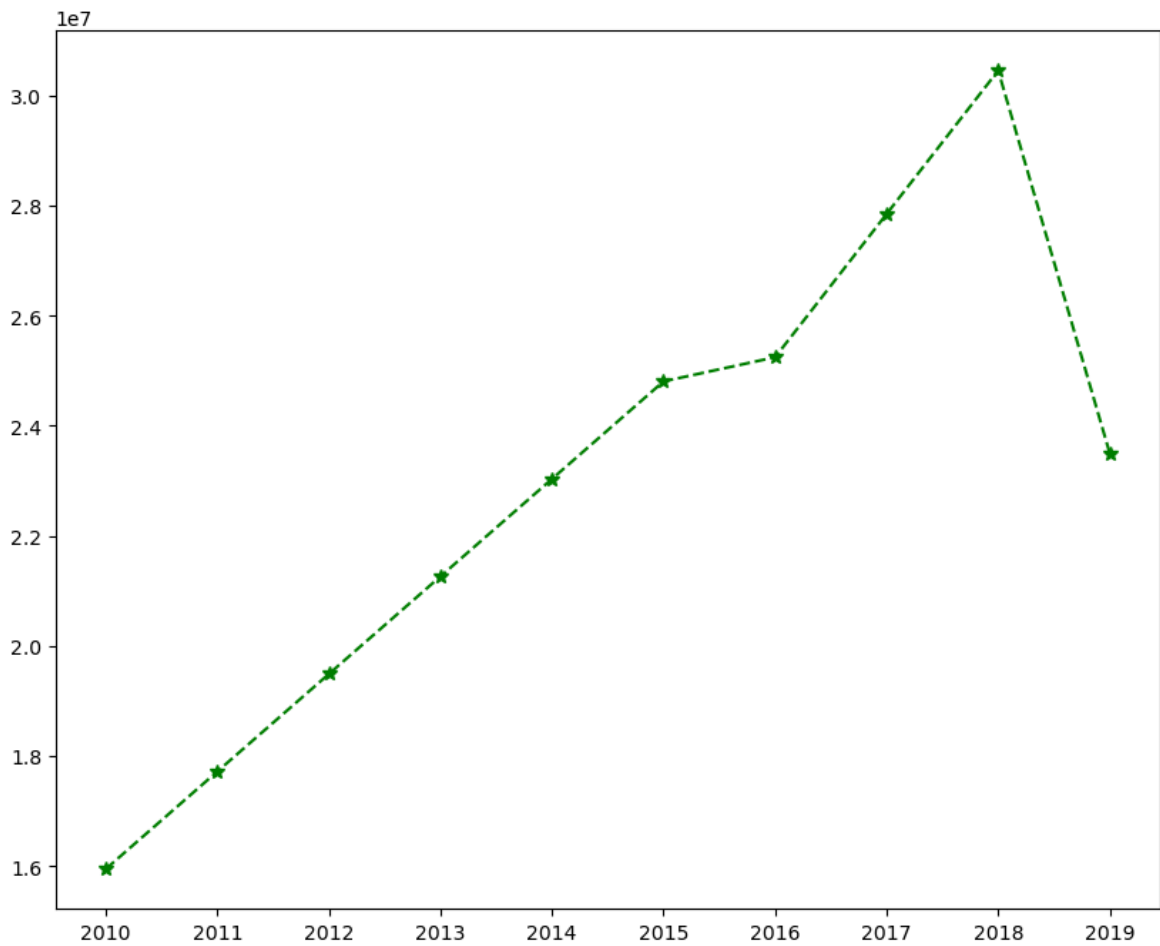
```
plt.plot(Salary[0], c='Green', ls = '-.', marker = '+', ms = 7, label = Players[0])  
plt.xticks(list(range(0,10)), Seasons, rotation='vertical')  
plt.show()
```



In [87]: Games

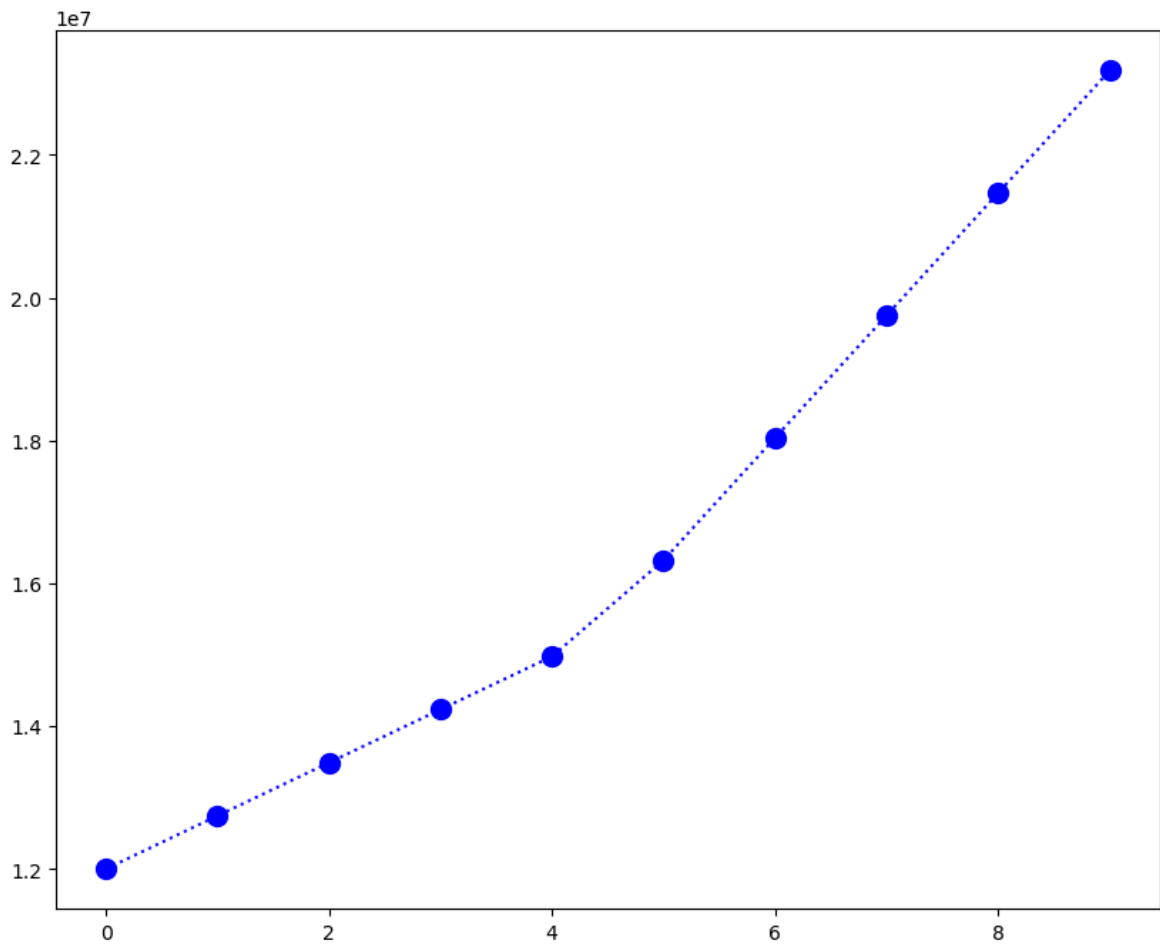
```
Out[87]: array([[80, 77, 82, 82, 73, 82, 58, 78, 6, 35],
                [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
                [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
                [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
                [82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
                [70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
                [78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
                [35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
                [40, 40, 40, 81, 78, 81, 39, 0, 10, 51],
                [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]])
```

```
In [134... plt.plot(Salary[0], c='Green', ls = '--', marker = '*', ms = 7, label = Players[
plt.xticks(list(range(0,10)), Seasons,rotation='horizontal')
plt.show()
```

```
In [89]: plt.plot(Salary[1], c='Blue', ls = ':', marker = 'o', ms = 10, label = Players[1])
```

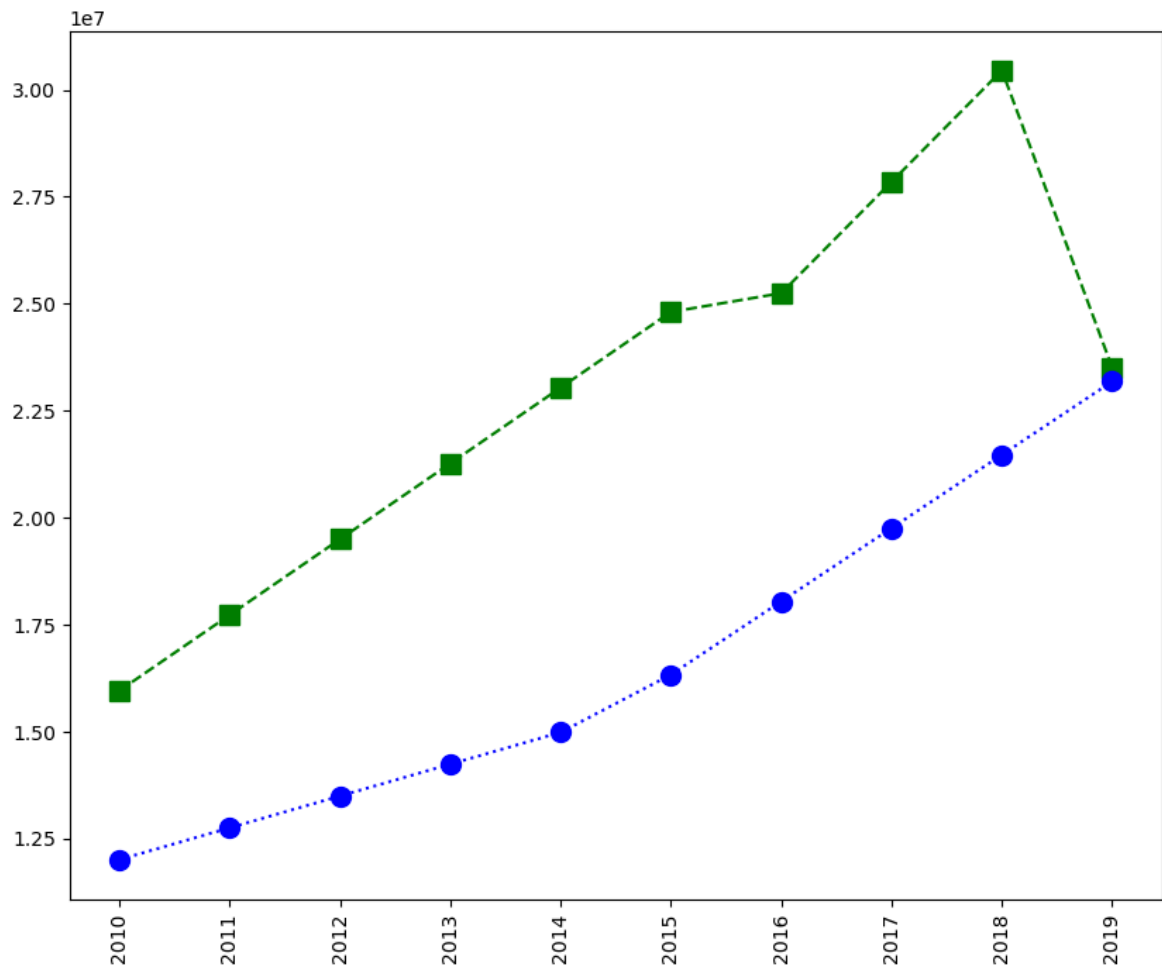
```
Out[89]: [<matplotlib.lines.Line2D at 0x1ef3c234b90>]
```



More Visulazation

```
In [92]: plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 10, label = Players[0])
plt.plot(Salary[1], c = 'Blue', ls = ':', marker = 'o', ms = 10, label = Players[1])

plt.xticks(list(range(0,10)), Seasons, rotation = 'vertical')
plt.show()
```

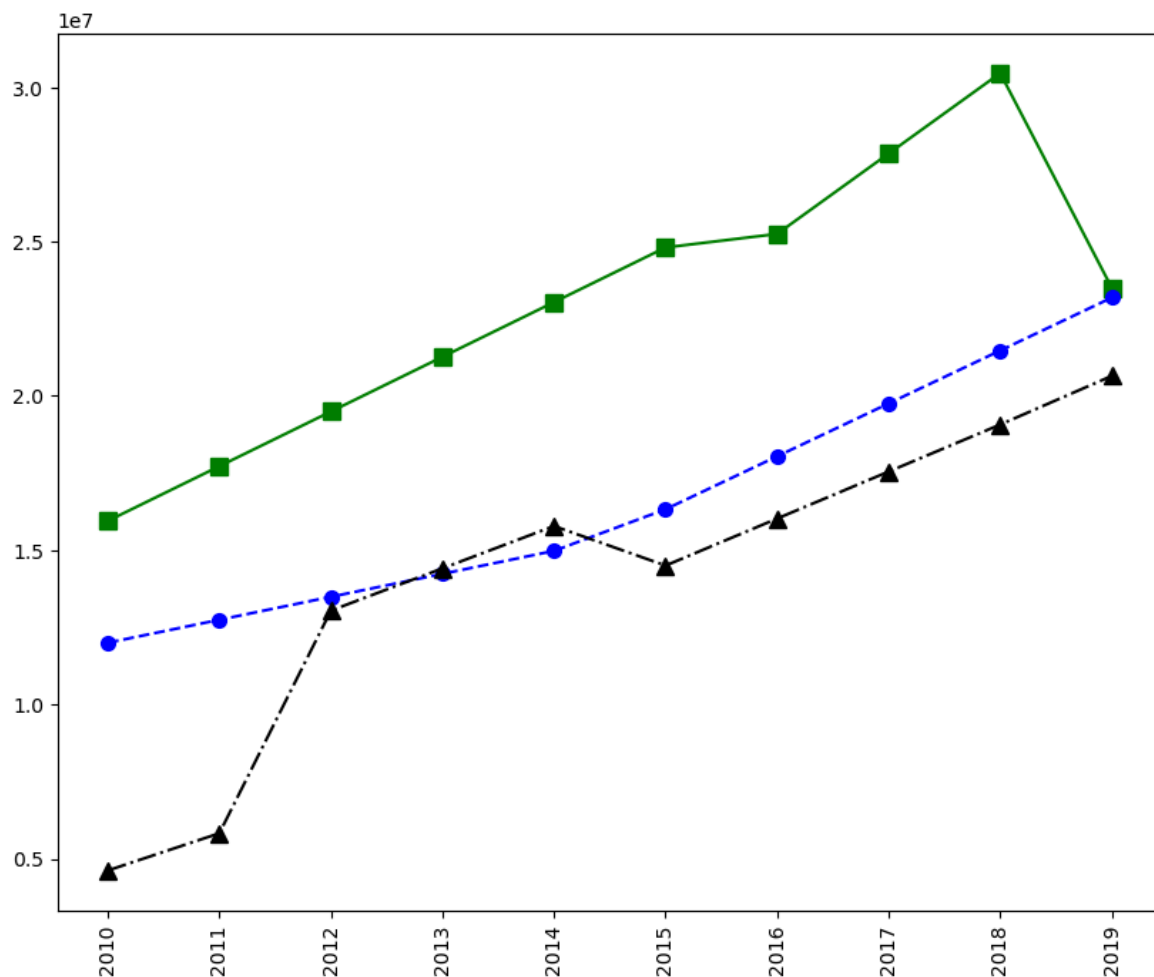


In [130...

```
plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 8, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '...', marker = 'o', ms = 7, label = Players[1])
plt.plot(Salary[2], c='k', ls = '-.', marker = '^', ms = 9, label = Players[2])

plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show() # Adding 3 pleyers with differant line style and size(ms)
```

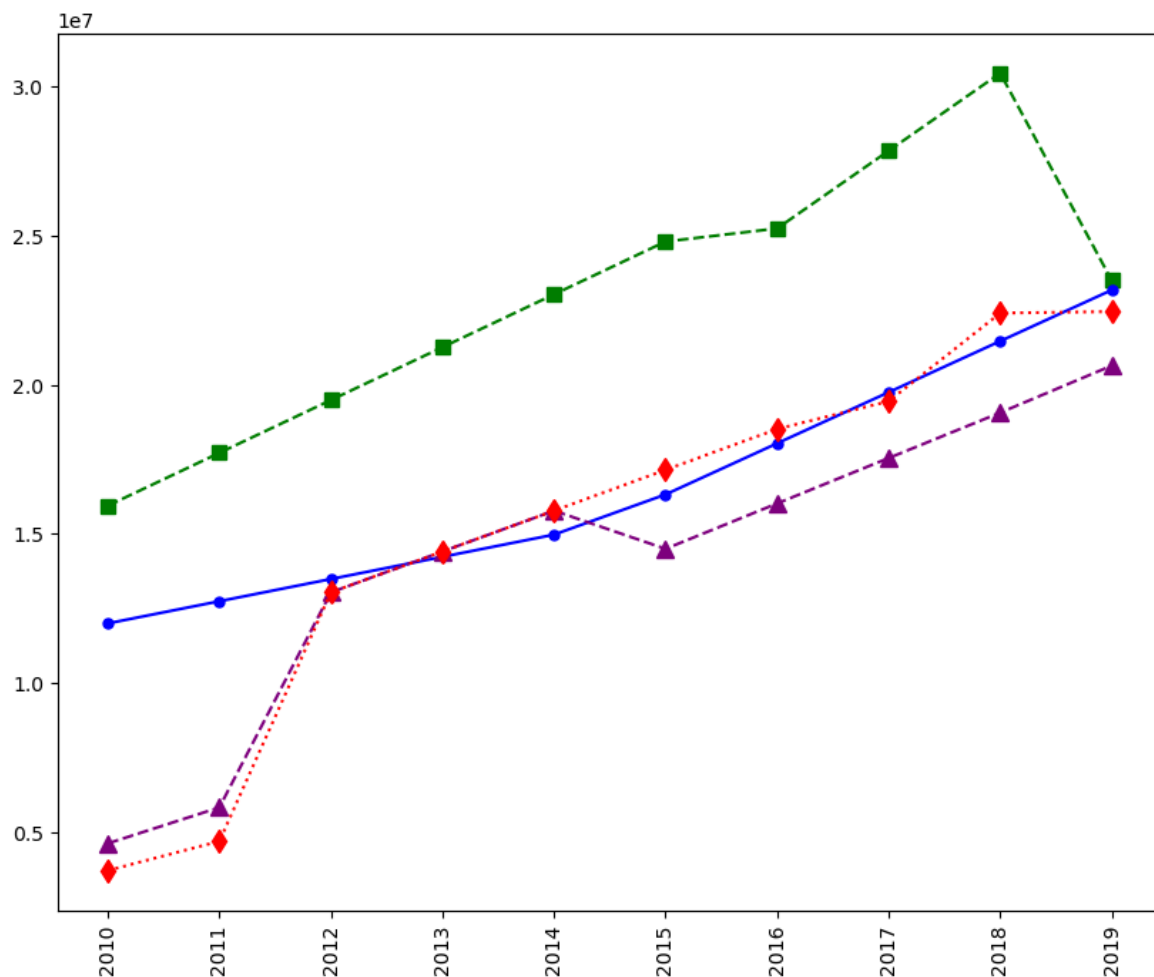


In [129...

```
plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '-', marker = 'o', ms = 5, label = Players[1])
plt.plot(Salary[2], c='purple', ls = '--', marker = '^', ms = 8, label = Players[2])
plt.plot(Salary[3], c='Red', ls = ':', marker = 'd', ms = 8, label = Players[3])

plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

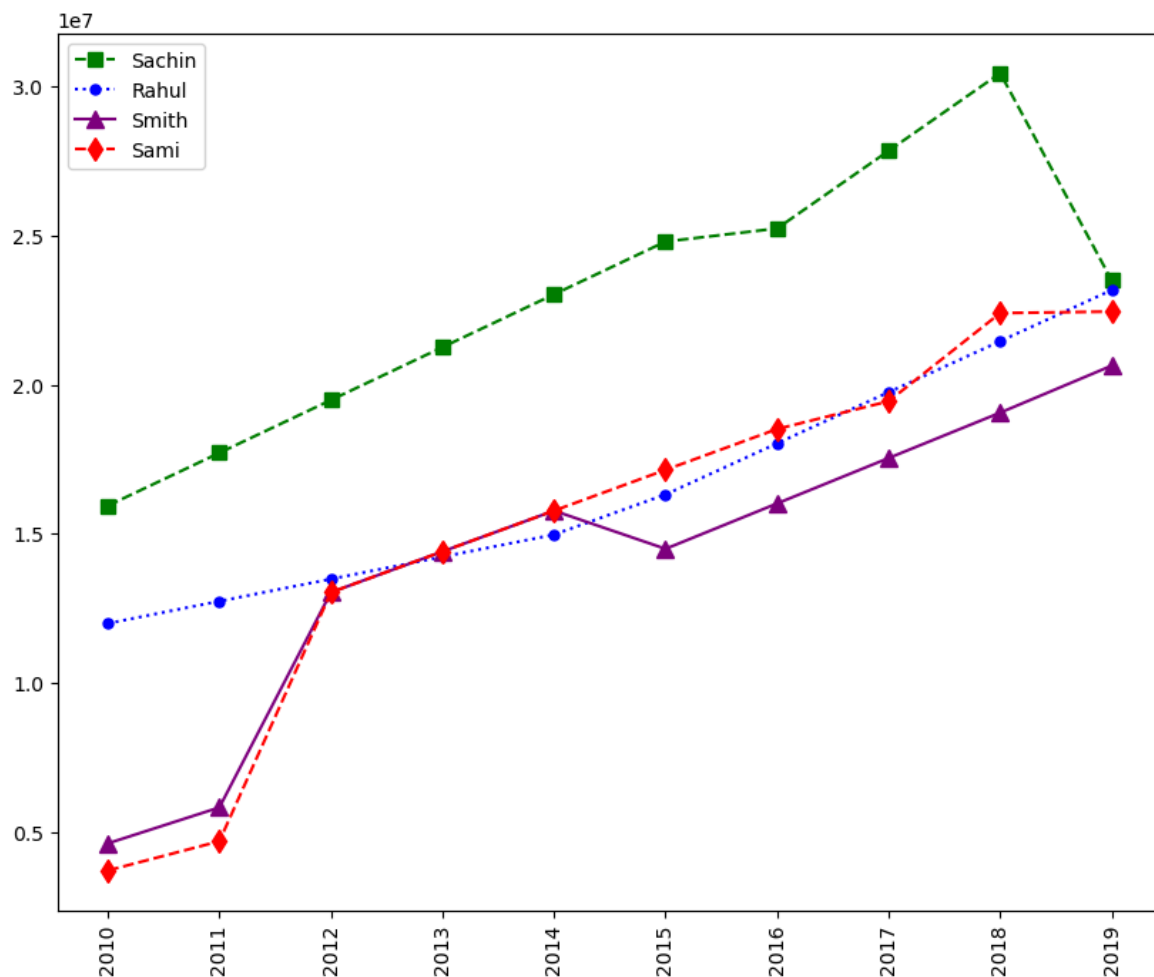
plt.show() # Here we add 4 plyers with differant color.
```



In [95]: *# how to add legend in visualisation*

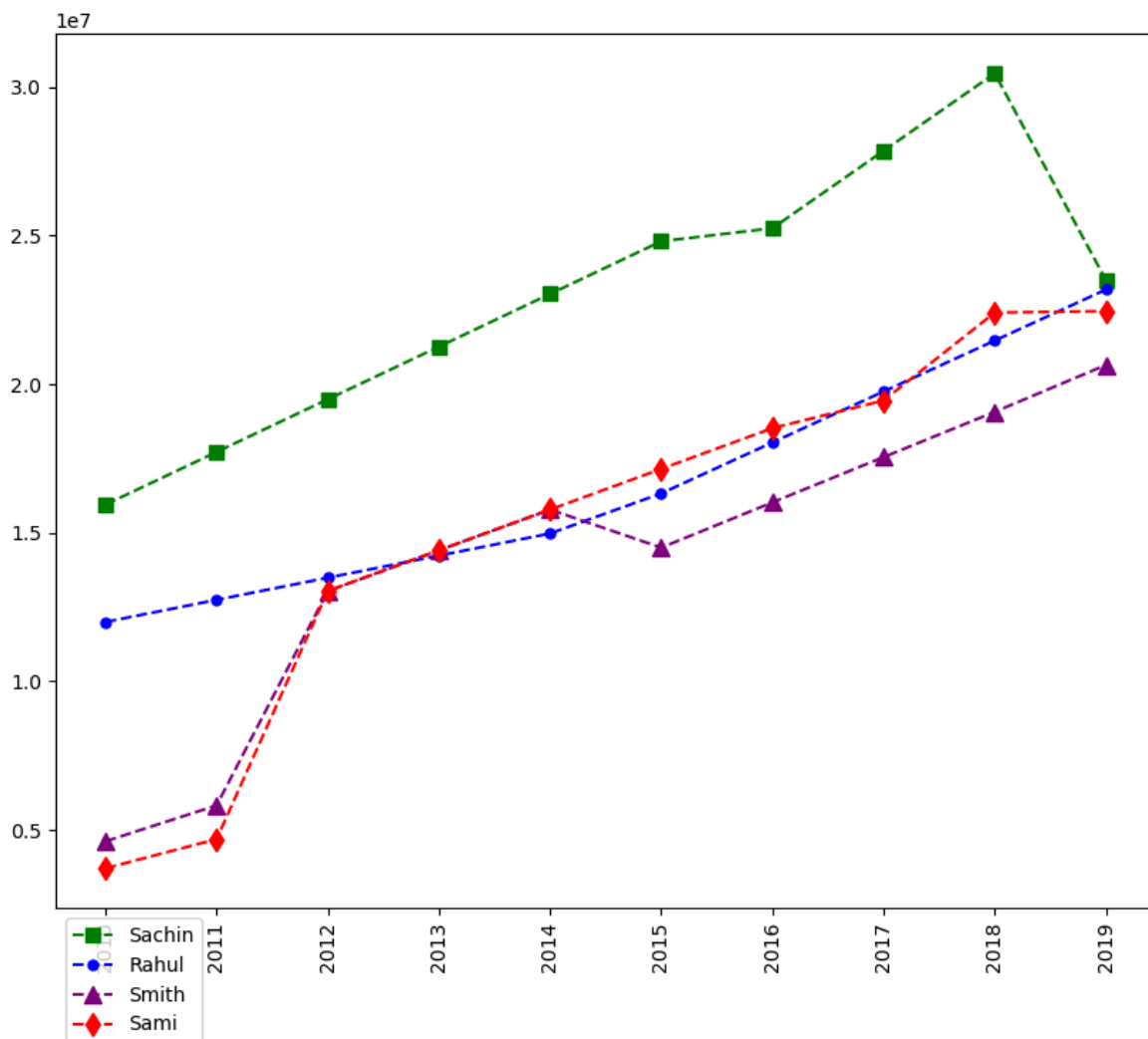
```
plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = ':', marker = 'o', ms = 5, label = Players[1])
plt.plot(Salary[2], c='purple', ls = '-', marker = '^', ms = 8, label = Players[2])
plt.plot(Salary[3], c='Red', ls = '--', marker = 'd', ms = 8, label = Players[3])
plt.legend()
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show()      # here our legend box is inside the graph
```



```
In [96]: plt.plot(Salary[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '--', marker = 'o', ms = 5, label = Players[1])
plt.plot(Salary[2], c='purple', ls = '--', marker = '^', ms = 8, label = Players[2])
plt.plot(Salary[3], c='Red', ls = '--', marker = 'd', ms = 8, label = Players[3])
plt.legend(loc = 'upper left',bbox_to_anchor=(0,0) )
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

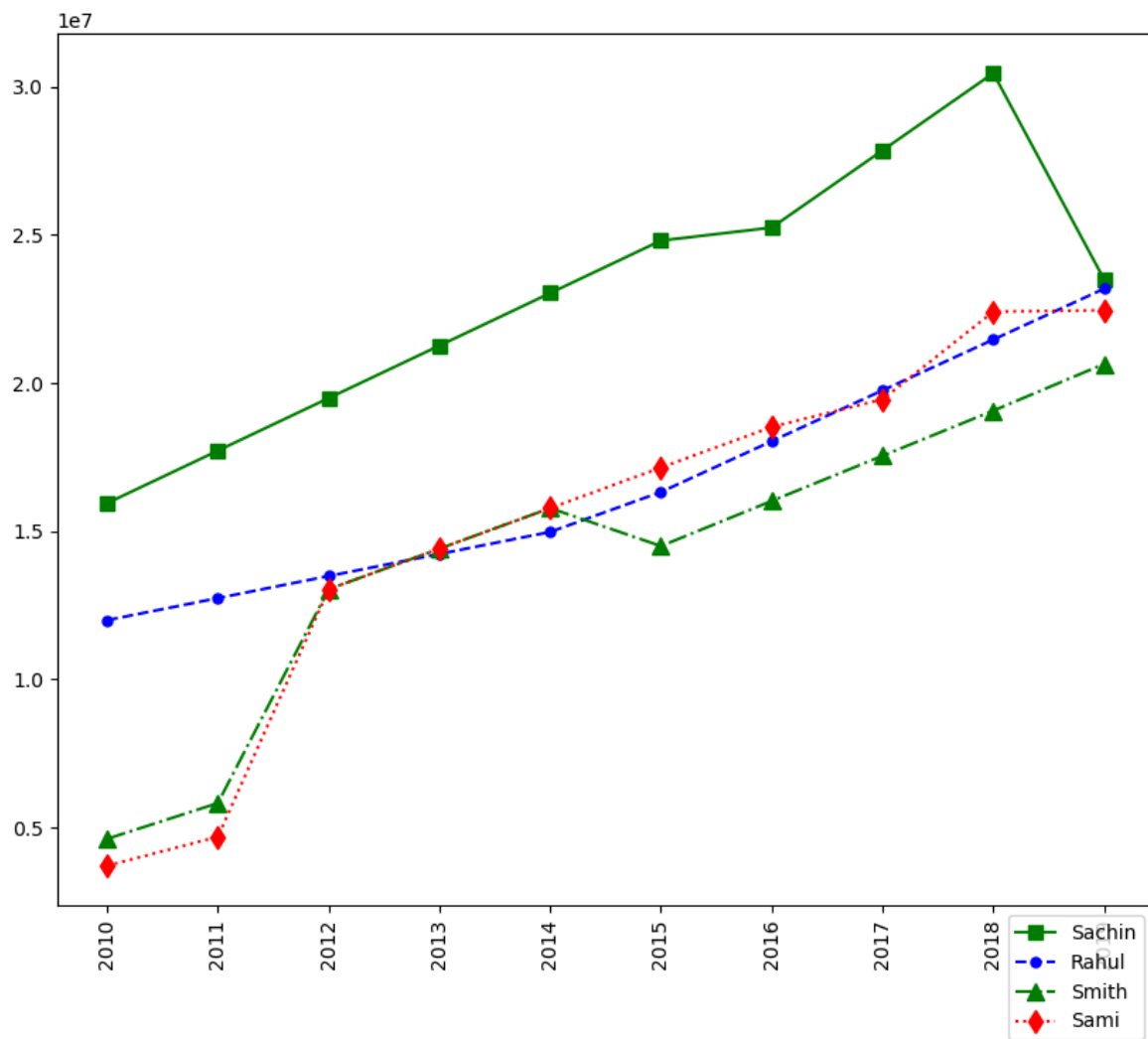
plt.show()    # Here our Legend box is outside the graph,but it show bottam Left
```



In [127...

```
plt.plot(Salary[0], c='Green', ls = '-', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '--', marker = 'o', ms = 5, label = Players[1])
plt.plot(Salary[2], c='Green', ls = '-.', marker = '^', ms = 8, label = Players[2])
plt.plot(Salary[3], c='Red', ls = ':', marker = 'd', ms = 8, label = Players[3])
plt.legend(loc = 'upper right',bbox_to_anchor=(1,0) )
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

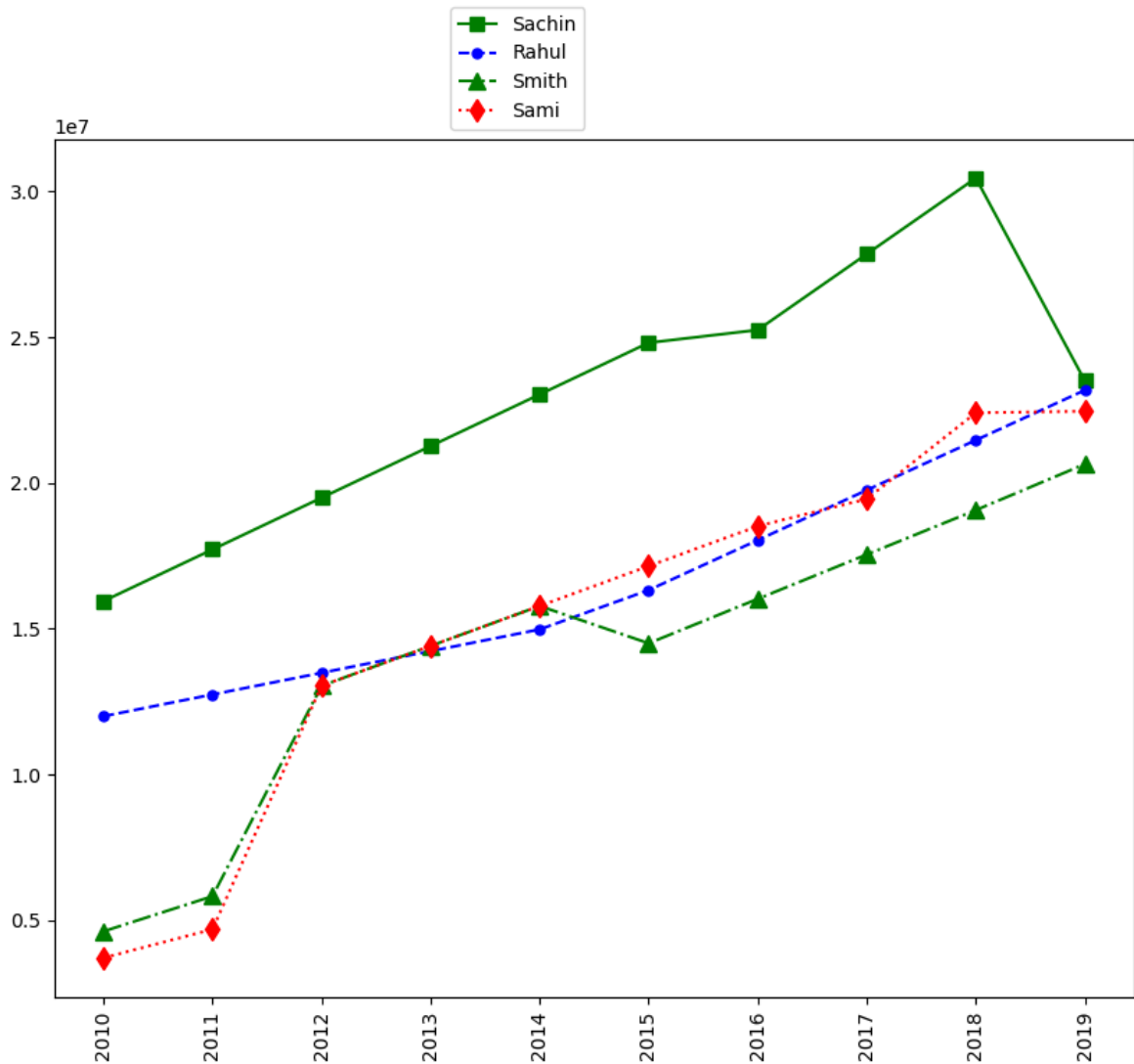
plt.show()      # Here our Legend box is outside the graph,but it show bottam ri
```



In [126...

```
plt.plot(Salary[0], c='Green', ls = '-', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '--', marker = 'o', ms = 5, label = Players[1])
plt.plot(Salary[2], c='Green', ls = '-.', marker = '^', ms = 8, label = Players[2])
plt.plot(Salary[3], c='Red', ls = ':', marker = 'd', ms = 8, label = Players[3])
plt.legend(loc = 'lower right',bbox_to_anchor=(0.5,1) )
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show() # Here our Legend box is outside the graph,but it show bottam right
```

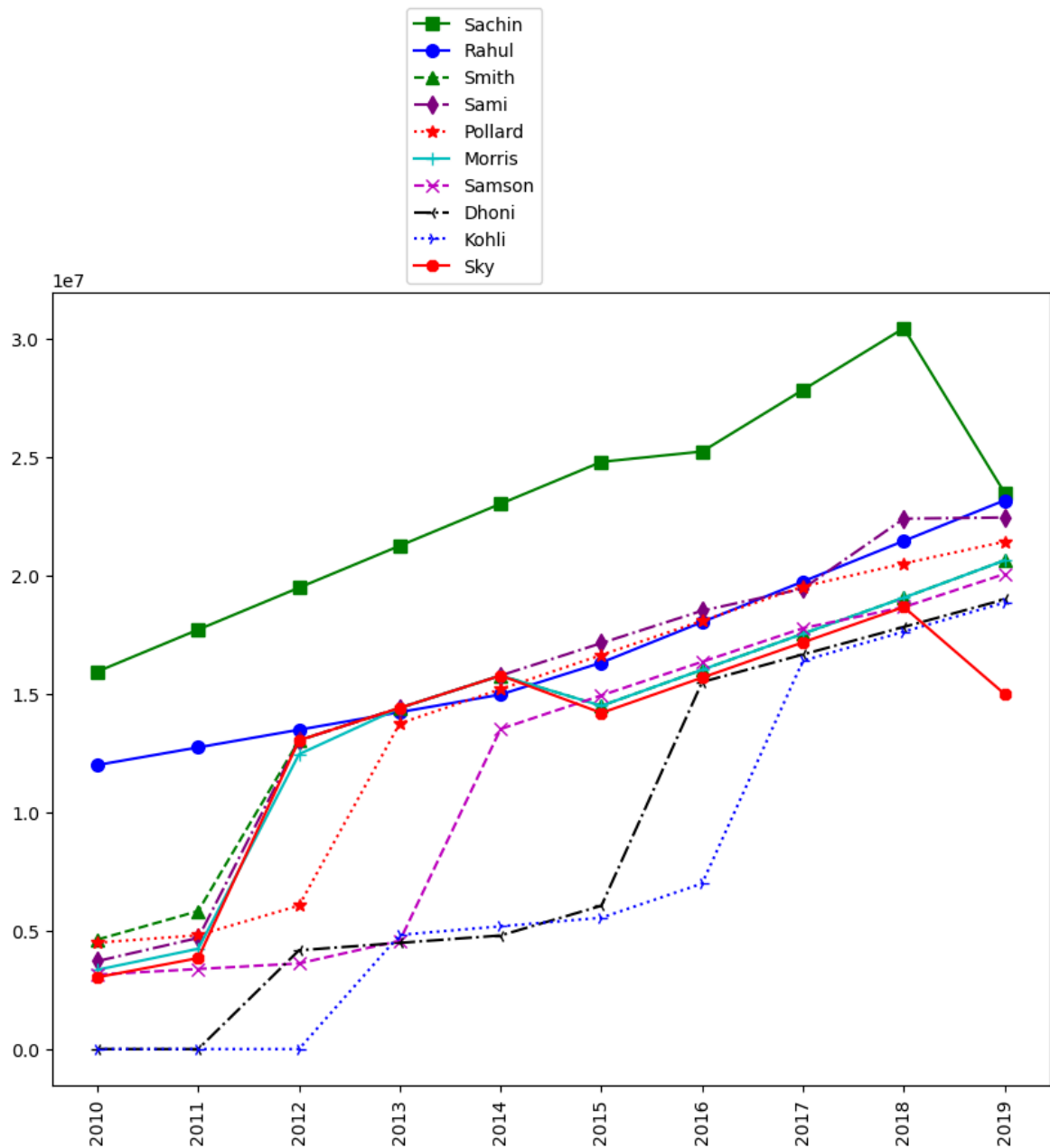



In [124...

```
plt.plot(Salary[0], c='Green', ls = '-', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '-', marker = 'o', ms = 7, label = Players[1])
plt.plot(Salary[2], c='Green', ls = '--', marker = '^', ms = 7, label = Players[2])
plt.plot(Salary[3], c='Purple', ls = '-.', marker = 'd', ms = 7, label = Players[3])
plt.plot(Salary[4], c='r', ls = ':', marker = '*', ms = 7, label = Players[4])
plt.plot(Salary[5], c='c', ls = '-', marker = '+', ms = 7, label = Players[5])
plt.plot(Salary[6], c='m', ls = '--', marker = 'x', ms = 7, label = Players[6])
plt.plot(Salary[7], c='k', ls = '-.', marker = '3', ms = 7, label = Players[7])
plt.plot(Salary[8], c='b', ls = ':', marker = '4', ms = 7, label = Players[8])
plt.plot(Salary[9], c='red', ls = '-', marker = '8', ms = 7, label = Players[9])

plt.legend(loc = 'lower right',bbox_to_anchor=(0.5,1) )
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show() # by this lower right command we get legend on top right side
```



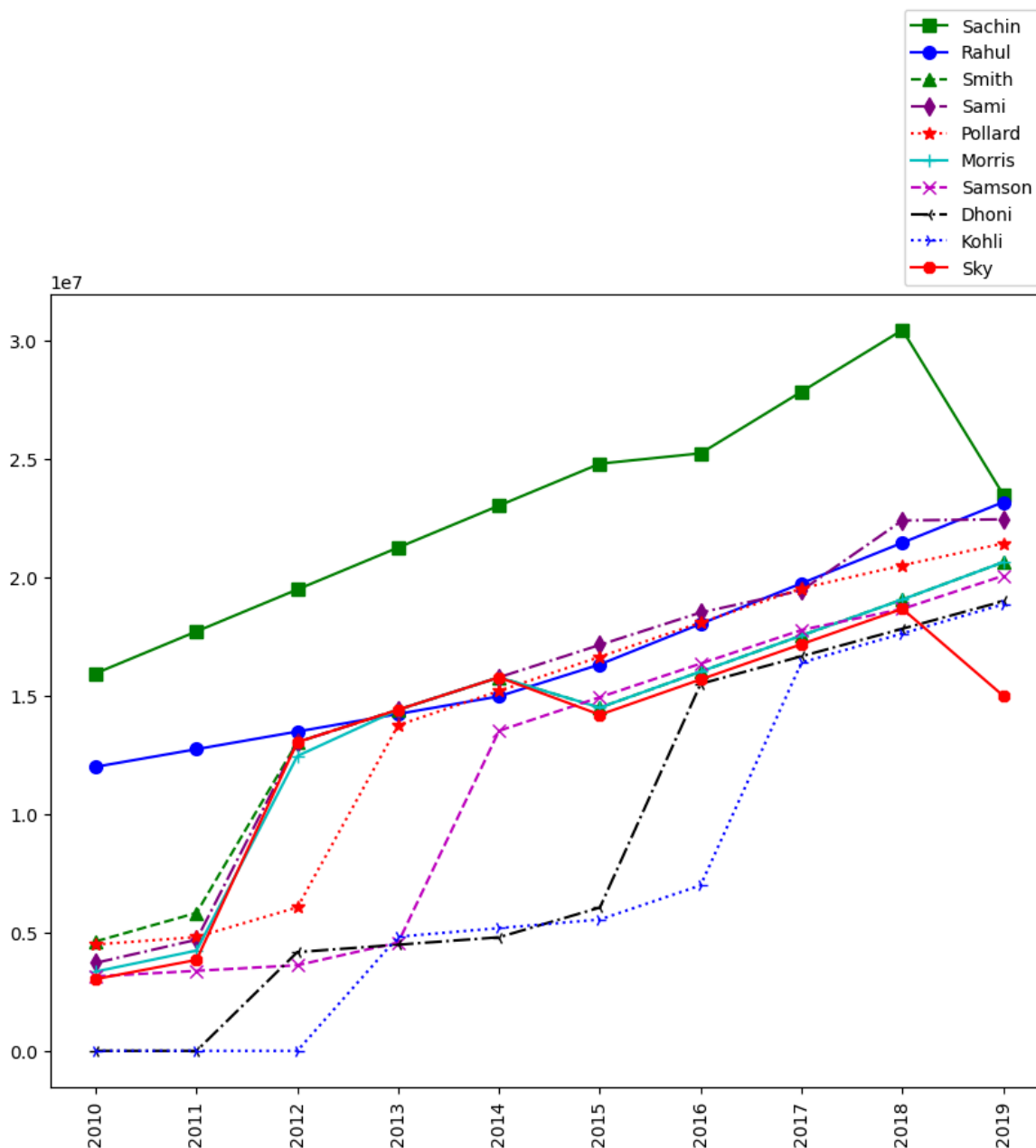
In [125...

we can visualize the how many games played by a player

```
plt.plot(Salary[0], c='Green', ls = '-', marker = 's', ms = 7, label = Players[0])
plt.plot(Salary[1], c='Blue', ls = '-', marker = 'o', ms = 7, label = Players[1])
plt.plot(Salary[2], c='Green', ls = '--', marker = '^', ms = 7, label = Players[2])
plt.plot(Salary[3], c='Purple', ls = '-.', marker = 'd', ms = 7, label = Players[3])
plt.plot(Salary[4], c='r', ls = ':', marker = '*', ms = 7, label = Players[4])
plt.plot(Salary[5], c='c', ls = '-', marker = '+', ms = 7, label = Players[5])
plt.plot(Salary[6], c='m', ls = '--', marker = 'x', ms = 7, label = Players[6])
plt.plot(Salary[7], c='k', ls = '-.', marker = '3', ms = 7, label = Players[7])
plt.plot(Salary[8], c='b', ls = ':', marker = '4', ms = 7, label = Players[8])
plt.plot(Salary[9], c='red', ls = '-', marker = '8', ms = 7, label = Players[9])

plt.legend(loc = 'lower right',bbox_to_anchor=(1,1) )
plt.xticks(list(range(0,10)), Seasons,rotation='vertical')

plt.show() # here by changing diamention of anchor, Legend completly goes at
```



• In this section we learned -

1) Matrices

2) Building matrices - np.reshape

3) Dictionaried in python (order doesnot mater) (keys & values)

4) visualizaing using pyplot

5 ipl data analysis

In []: